

# EXECUTIVE SUMMARY

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## Case Overview

The Cook County Assessor's Office (CCAO) is responsible for accurately testing property values to determine fair taxation within Cook County, Illinois. However, CCAO's valuation methods lacked transparency and accuracy. It causes significant public and political pressure. Therefore, since 2018, the office has modernized its processes, emphasizing transparency and data-driven modeling.

As part of this modernization effort, the objective of the project is to build a predictive model for estimating the market value of residential properties using provided datasets. The historical and predicted property data collected from CCAO. The historical data contains 50000 records of historical property variables. The predicted data contains 10000 records. The primary goal of our team is to minimize the Mean Squared Error (MSE) between predicted property values (modeled) and actual market values, ensuring assessments are both fair and accurate.

As a data scientist on this team, we will do the following process to increase the accuracy of the model:

1. Analyzing historical property sale price data to identify potential patterns and trends.
2. Eliminating the null value if 10% of the column data are null values and delete useless columns.
3. Selecting a machine learning method to build a model to predict the sale price of properties.
4. Building, testing, and optimizing predictive models for property valuation.
5. Generating actionable insights and recommendations to further improve the accuracy and reliability of property assessments.

## Methodology

Describe the data approach and methodology you use. Justify your choice of methodology. You should be precise but avoid jargon. This is a document that will potentially be shared with the data science team at CCAO, so make sure the language is clear, professional, and appropriate for them.

### Pre-select variables

At the beginning phase of building the model, our team should make sure that all 62 variables were suitable to apply in the predicted property price.

Firstly, the data was refined by dropping variables if the null value was more than 10% of the data (over 5000 we do not want to sacrifice the number of records more than 5000 to keep a variable. Though we can change the missing value to a consistent value (mode or mean), it will increase the data bias) and we deleted the columns that were not identified as good predictors based on the codebook.

After checking the codebook and dropping the variables listed as not important, it left us with 41 variables. Later, we should make sure the datatype (numeric or categorical) was corrected. This comprehensive pre-selecting variable made sure the dataset was ready to establish the model.

## Models Execution

### 1. Linear regression

We attempt to utilize the linear regression model to predict the sale prices of properties.

Firstly, before using variable selection methods, our team created a correlation matrix to visualize the correlation between explanatory variables and dropped the ones with a correlation higher than 70%. Afterward, our team applied a **forward stepwise method to filter important variables for linear regression**.

To make sure the stepwise method would not miss any possible, we used lasso regression to shrink insignificant variables to zero for linear regression to make sure the model had been utilized. Finally, we randomly separated the training and test datasets and got an MSE (mean of squared error) of **15,802,894,150**.

### 2. Lasso regression

We directly applied variables that can be identified by Lasso Regression (Any variables with coefficients exactly equal to 0 are excluded, indicating they are not important predictors under the Lasso penalty). It resulted in an MSE (mean of squared error) of **15,812,930,211**.

### 3. Random Forest

Since Random Forest accounts for non-linear relationships, we decided not to use variables filtered by Lasso or matrix correlation. Instead, our team opted to reselect variables based on their importance as determined by Random Forest.

Initially, we considered Recursive Feature Elimination (RFE), but its computational complexity and inefficiency led us to choose a different approach. Using the Random Forest ranger model in R (**with mtry set to 6 and num.trees to 150**), we identified the most important variables. From the importance rankings, we selected 32 variables to construct a refined Random Forest model. To mitigate overfitting, we set **mtry to 5 and num.trees to 500** in this second model.

The final model achieved a Mean Squared Error (MSE) of **11,140,583,987.622**, which is lower than the MSEs obtained from linear and Lasso regression models.

## Summary method

After comparing three models—linear regression, Lasso regression, and Random Forest—our team selected Random Forest as the final model for predicting property sale prices, as it achieved the lowest mean squared error (MSE).

## Conclusion

Describe your results, including summary statistics (e.g., min, max, mean, and quartiles) of the distribution of assessed property values. Describe your data file which reports the assessed property values you have generated.

We analyzed the frequency histogram of the prediction results and observed that the predicted values exhibit a right-skewed normal distribution. The predicted price range spans from a minimum of **18,526.7** to a maximum of

**326,849,9.8**, indicating a significant spread. Most of the predicted prices are concentrated in and around the range of **156,526** to **202,526**, where the frequency is highest. This highlights the clustering and variability within our data.

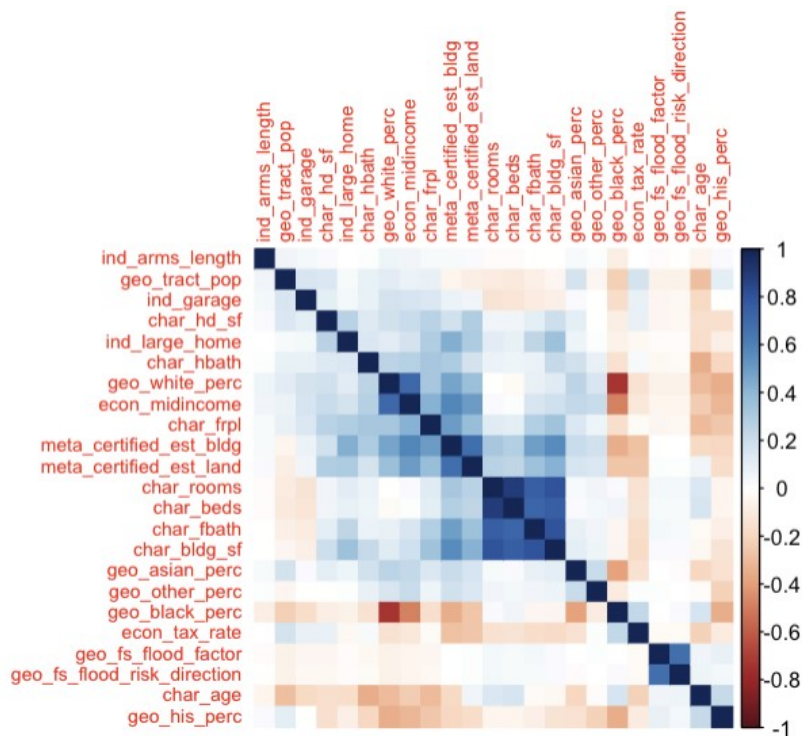
Further analysis reveals that the average predicted price is **320,937.3**, which falls outside the primary concentration range, reinforcing the right-skewed nature of the distribution. Additionally, the first quartile (Q1) is **152,488.8**, and the third quartile (Q3) is **380,740.1**, meaning nearly 50% of the predictions lie within this interval.

In summary, the predicted values demonstrate a right-skewed distribution with a large gap between the maximum and minimum values, emphasizing the asymmetry and dispersion within the dataset.

## Appendix

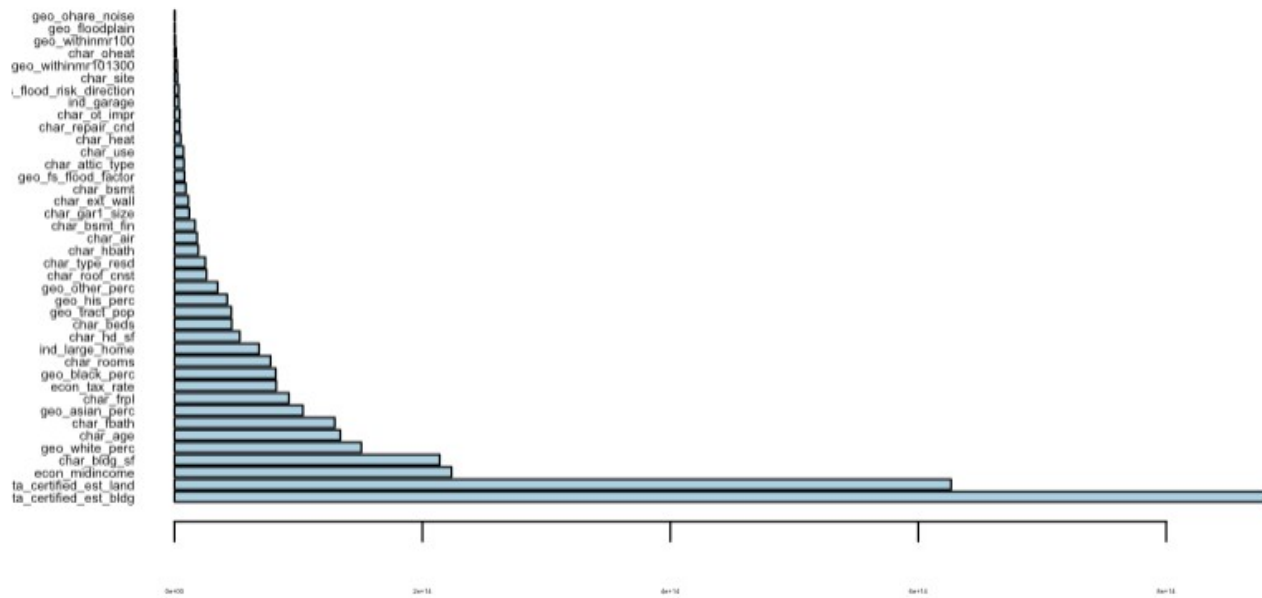
Include tables and plots here. Enumerate each table and plot, give each a descriptive title, and make sure elements are labeled clearly. Tables and plots should correspond to output from your code. 1.

Correlation Matrix in linear regression



## 2. Variable Importance under the method of Random Forest

### Variable Importance



### 3. Assessed Variable Distribution

|              |           |
|--------------|-----------|
| min          | 18526.7   |
| max          | 3268499.8 |
| mean         | 320937.3  |
| 1st quartile | 152488.8  |
| 3rd quartile | 380740.1  |

